PortableTAK and the Skydio X10D

Put the aircraft on the TAK map and share its live video with every TAK user.

When you finish, every ATAK, iTAK and WinTAK user connected to PortableTAK sees the X10D on the map and can open its live video. Allow about 20 minutes the first time. Menu names on the controller can differ slightly between Skydio software versions.

Before you start

You need	Details
Skydio X10D and controller	Updated, with the licensing that enables **ATAK Services**, **RTSP Services** and **Wi-Fi** (Skydio Support enables these on your Skydio cloud account).
PortableTAK running	On an OpenMANET node, a Windows computer or a Mac, and the dashboard open as administrator.
A USB-C flash drive	Formatted **exFAT** (the controller does not read NTFS or FAT32).
Internet on the server (hotspot setup only)	Plus **Remote Access** set up once in the dashboard (see the Setup and Troubleshooting Guide).

1. Choose how the controller reaches the server

Setup	When	Server address to use below
A. Same network	The controller joins the same Wi-Fi as the server: the node's Wi-Fi, or the Wi-Fi the Windows/Mac computer uses.	The server's address for phones (node: `10.41.250.10`)
B. Phone hotspot bridge	The drone team is on cellular and the server (e.g. a Windows laptop) is somewhere else. The controller joins the hotspot of an ATAK phone (e.g. the S22), and that phone forwards the drone's traffic to the server.	The phone's **hotspot address** shown by the bridge (e.g. `192.168.43.1`)

Setup B: start the phone bridge

Nothing is installed on the controller. The phone uses its own Tailscale to reach the server, so the server needs **Remote Access** set up and internet, and the phone needs the Tailscale app (signed in, switched on).

1. On the phone, once: install **Termux** (free, from Google Play or F-Droid).
2. Dashboard (open it on the phone over Tailscale) → **Video Feeds** → **Drone far from this server (phone hotspot bridge)** → **Copy command**.
3. Turn on the phone's **hotspot** and **Tailscale**. Open Termux, paste the command, press Enter. It prints the **drone address** (the hotspot line, e.g. `swlan0: 192.168.43.1`). If none shows, use the **Gateway** in the controller's Wi-Fi details.

4. Join the controller to the phone's hotspot. Use the drone address as the server IP in steps 3 and 5 below.
5. Leave Termux open while flying (Android may stop it if closed). Run the same command again any time; it replaces the old bridge.

> Video over cellular uses a lot of data on the phone's plan.

2. Make the aircraft's certificates (dashboard)

1. Dashboard → **Users** → create a user for this aircraft, for example `skydio1`. Use one user per aircraft.
2. On that user, press **Device TLS Package**. It downloads `skydio1_TLS.zip`.
3. Unzip it and copy its files to the USB-C drive:

File	What it is
`truststore-root.p12`	The Trust Store: lets the controller trust this server
`skydio1.p12`	The aircraft's client certificate: proves it to the server
`secure.pref`	The connection settings and certificate passwords (same layout as a Sit(x) TLS package)
`README.txt`	Every value to type on the controller, including both passwords

> Keep the package and the USB-C drive private: they contain the certificate passwords. Delete the files from the drive when you are done.

3. Connect the controller to PortableTAK (aircraft on the map)

1. Turn on the X10D and the controller. Setup B: start the phone bridge first and join the controller to the phone's hotspot.
2. Plug the USB-C drive into the controller.
3. **Global Settings** → **Sharing** → **TAK/CoT** → **TAK Connections** → **Add a Connection**.
4. Enter the values from `README.txt`:

Field	Value
Server name	One word, as in the README (e.g. `TestaTAK`)
Server IP address	Setup A: the server's address. Setup B: the drone address from the bridge.
Server port	**8089**

5. **Advanced Options** → **SSL/TLS** → **Configure Trust Store**: pick `truststore-root.p12` on the USB-C drive and enter the **Trust Store password** from the README. If the controller also asks for a client certificate, pick `skydio1.p12` with the **client certificate password**.
6. Back → **Add Connection**. Remove the USB-C drive.
7. Check: in ATAK on a phone, the aircraft appears on the map once the controller is connected.

4. Add a video feed (dashboard)

1. Dashboard → **Video Feeds** → type a feed name → **Add feed**. Use lowercase letters, numbers and dashes, for example `skydio1`.
2. The card shows the exact settings for the controller, with this server's address. Press **Drone settings** to see them again later.

5. Stream the video (controller)

1. Menu (three lines, top left) → **Sharing** → **RTSP** → **External**.
2. Enter:

Field	Value
Stream name	The feed name, exactly (e.g. `skydio1`)
IP address	Setup A: the server's address. Setup B: the drone address from the bridge.
Port	**8554**

3. **Start Stream**. Video begins once the aircraft takes off. The dashboard's Video Feeds card then shows a green **Live** dot.

6. Watch the video

- **ATAK / WinTAK:** open **Video** and pick the feed by name.
- **Add it by hand** (any RTSP player): `rtsp://<server address>:8554/skydio1`. No user name or password is needed for drone feeds. In the dashboard, **Watch** plays it in the browser.
- ATAK users on cellular (Tailscale on) watch through the server's Tailscale address, wherever the server is.

Troubleshooting

Problem	What to do
Aircraft not on the map	Check the controller's TAK connection: port 8089, SSL on, both certificates and passwords from the README. Setup B: Termux still open with the bridge running, the phone's Tailscale on, and the server online.
Video Feeds never shows Live	Same stream name on both sides, port **8554**, the right IP for your setup, and the aircraft flying. Setup B: the bridge running on the phone, the phone's Tailscale on, and internet on the server.
Never shows Live while a TAK ICU feed exists	With a TAK ICU feed on the server, RTSP video is accepted over TCP only. If the controller cannot connect then, delete the TAK ICU feed (Video Feeds) while flying.
Video stops after switching color/thermal	Expected on the X10D: press **Stop Stream**, then **Start Stream**. Viewers reopen the feed.
Video choppy or delayed	Move closer to the Wi-Fi, lower the stream resolution or frame rate (30 FPS), or use a wired USB-C Ethernet adapter on the controller.
The server moved to another node or computer	Setup B keeps working: the bridge follows the drive's Tailscale address. Setup A: use the new server's address in both controller settings.
The phone's hotspot address changed	Run the bridge command again and use the address it prints in both controller settings.
A second aircraft	Repeat with a new user and feed name (e.g. `skydio2`).